Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

Team name: SKUdesk

Team leader name: Mr. Chonlarnatt Poonchattakarn

Team member: Ms. Ploypitcha Boontavipitak

Problem statement: Shoppers and small resellers cannot tell whether a cheaper listing is the identical product, so a “deal” is often a different model, a case or a clone. SKUdesk finds the cheapest landed cost, prices a resale and checks arbitrage only on provably identical products.

Brief about the idea

- SKUdesk is one agent for three jobs: find the cheapest landed cost, price a resale, and check arbitrage.
- It works only when two listings are the identical product. Gemini decides identity: it must quote the brand and model it relied on, and code checks those quotes are in the listing. A second Gemini step looks at the photo.
- The answer is often “no”: no identical product, or no gap left after fees, returns and postage. That is the point. A wrong “yes” loses money.
- The agent never spends money. Every buy is a proposal that a person approves. An approved resale can then be listed in Google Merchant Center by the owner.
- Real services end to end: Google Shopping results, Gemini, Cloud Run, Firestore, Cloud Scheduler, Secret Manager and the Merchant API, live at skudesk.lol.

Opportunities

- **How different is it?** Price trackers compare titles and show a “lowest price” that is often another model, an accessory or a clone. SKUdesk makes identity the first gate and shows every listing it turned down and why.

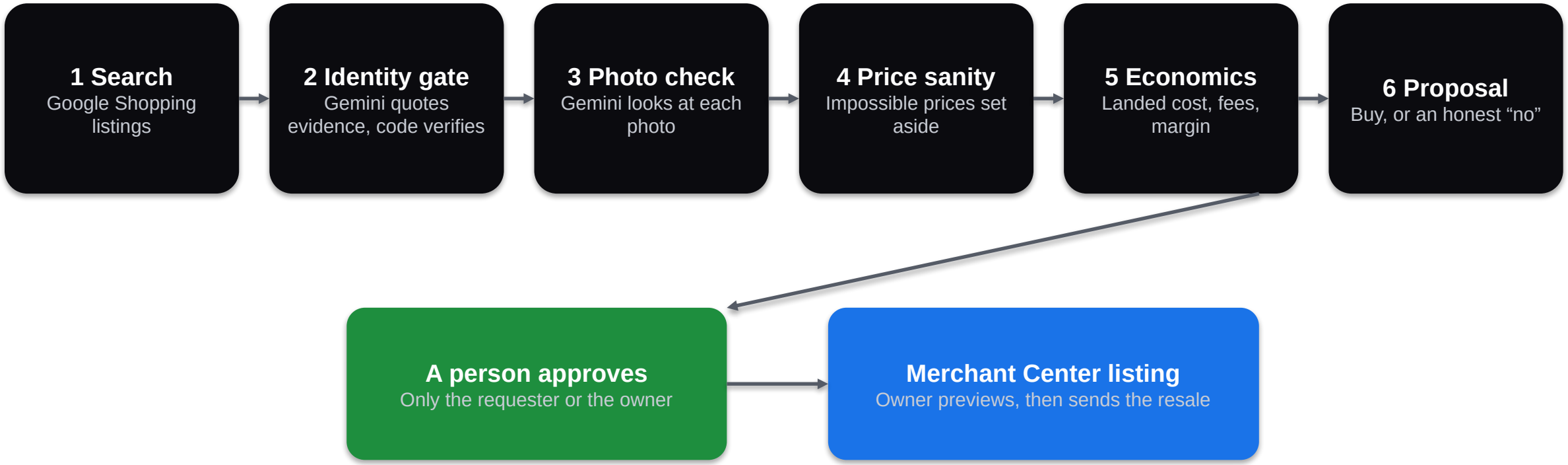
- **How does it solve the problem?** A two-step identity check (text with quoted evidence, then the photo), a price-sanity filter for impossible prices, and a cost model in integer cents: landed cost, 13% marketplace fee, 2% returns reserve, outbound postage and a margin floor.

- **USP** The honest “no”. 67 labelled product pairs ran against the real model with zero wrong matches, and a person approves every spend. A market view shows asks, four buyer policies’ bids and the spread, so the reason for a “no” is visible.

List of features offered by the solution

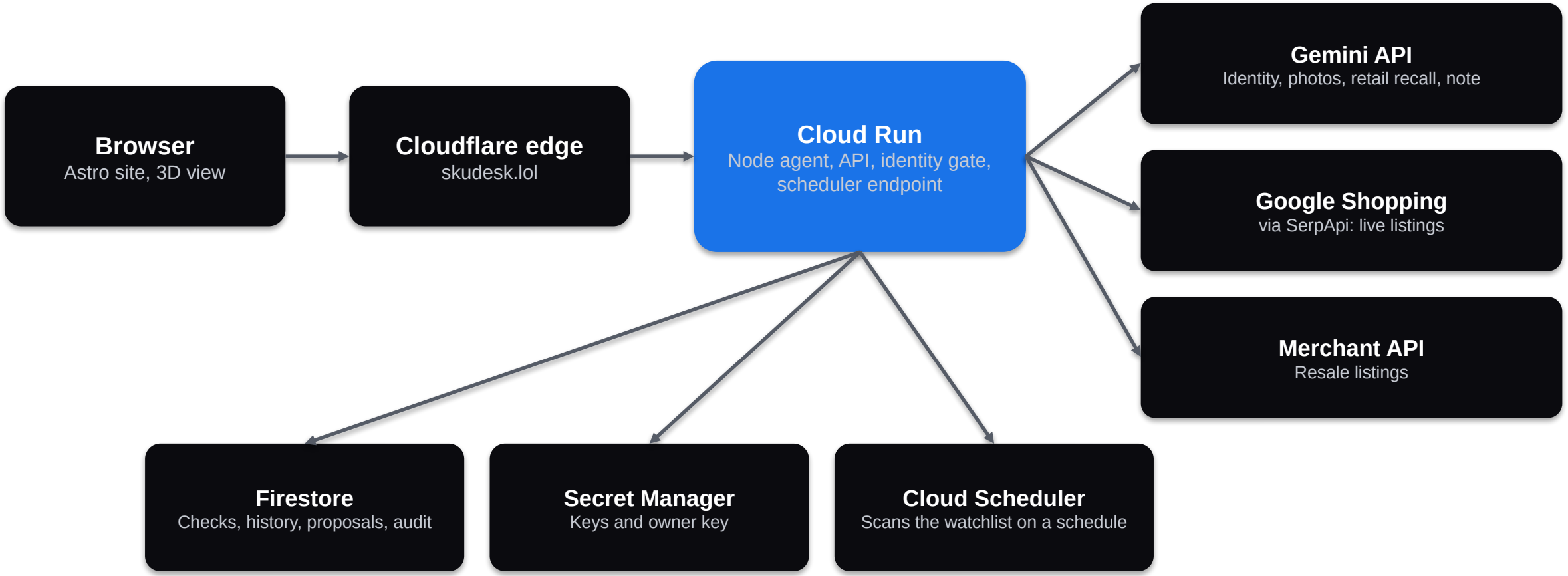
- **Identity gate** Gemini quotes brand and model; code verifies the quotes and reads storage size and model tier itself.
- **Photo check** Gemini compares each matching photo with the requested product and can only take a match away.
- **Price sanity** Prices far below the median or recalled retail are set aside as bad data.
- **Economics** Landed cost, fees, returns, postage and margin floor, in integer cents.
- **Approval** Every buy is a proposal; only the requester or the owner decides.
- **Market page** Order book of identical asks, four buyer-policy bids, spread, history chart, 3D product view.
- **Watchlist and scheduler** Up to 12 products checked on a schedule, longest-waiting first.
- **Live agent view** A graph that lights up as the agent searches, judges, looks and prices.
- **Resale listings** Preview and send an approved resale to Google Merchant Center (owner only).
- **Owner controls** Per-buy limit, margin floor, kill switch, audit log.

Process flow diagram



Nothing identical, or no gap after costs: the agent says no and shows what it turned down. The scheduler runs the same flow on the watchlist.

Architecture diagram of the proposed solution

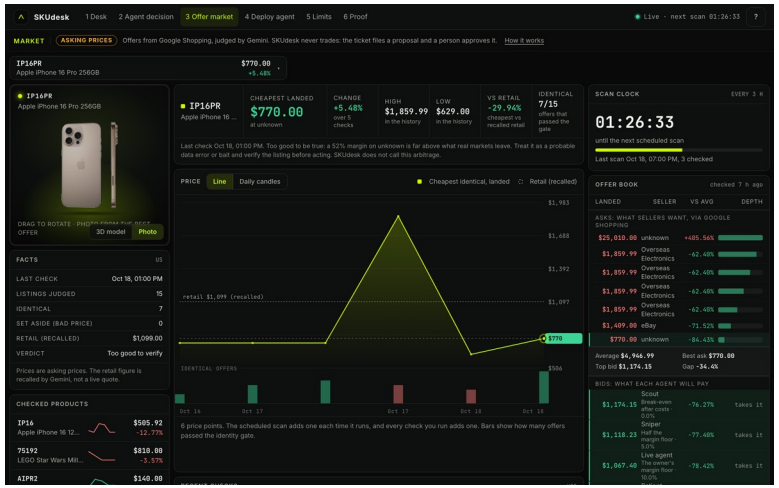


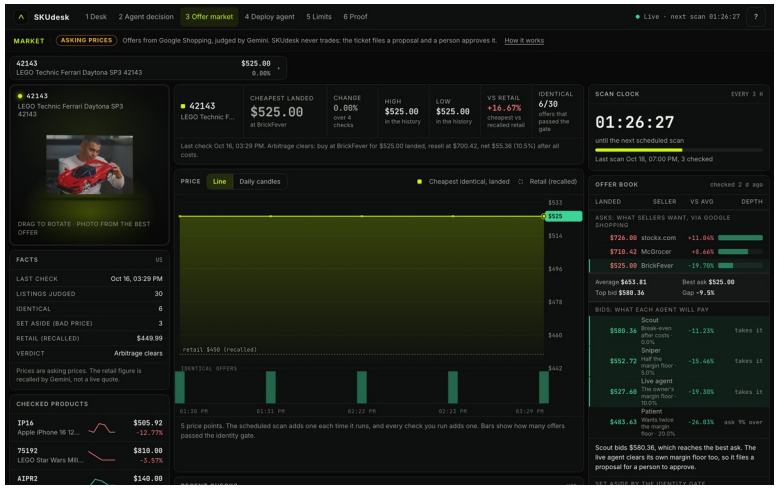
The edge worker adds a shared secret so the service only answers traffic that came through it. The page content policy allows scripts from the site only.

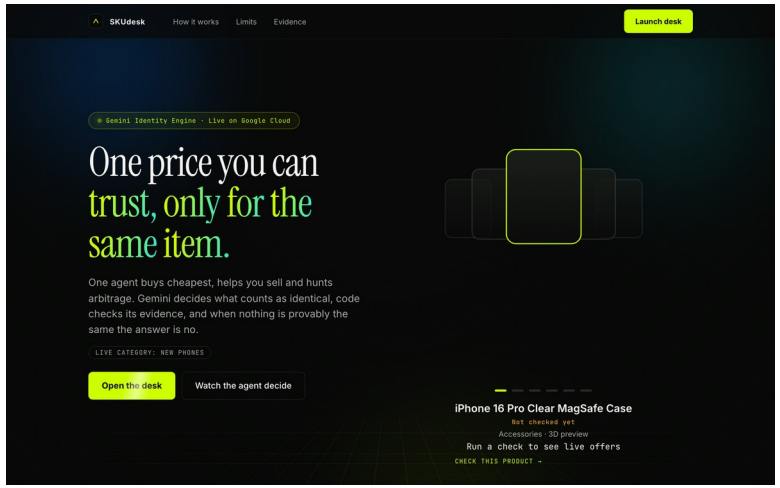
Technologies used in the solution

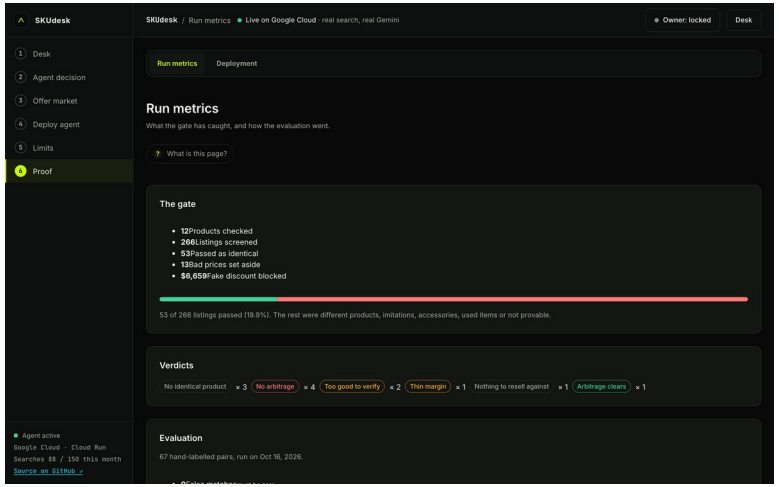
- **Google Cloud** Cloud Run, Firestore, Cloud Scheduler, Secret Manager, Merchant API
- **Gemini** Google Gen AI SDK: structured identity verdicts, image understanding, grounded retail recall
- **Google Shopping** Live listings through SerpApi
- **Backend** Node.js with native TypeScript, no framework, node:test
- **Frontend** Astro, vanilla TypeScript, WebGL 3D view, canvas
- **Quality** 136 unit tests, 21 browser end-to-end tests (Playwright), a labelled evaluation set, a smoke test for the deployed service

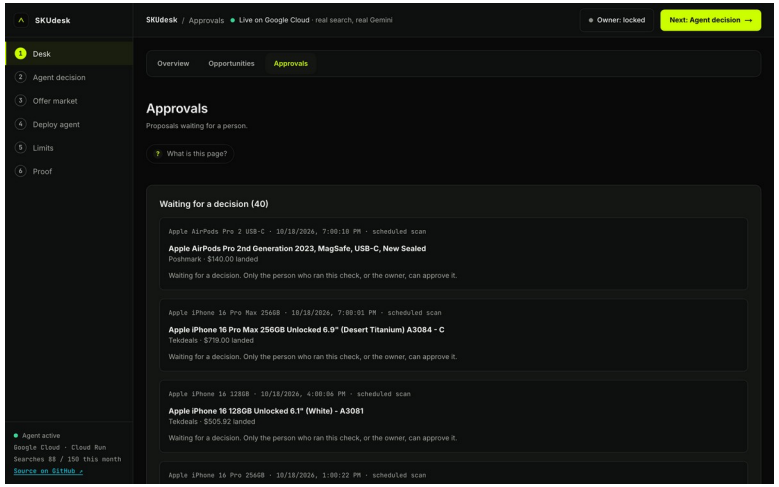
Snapshots of the prototype

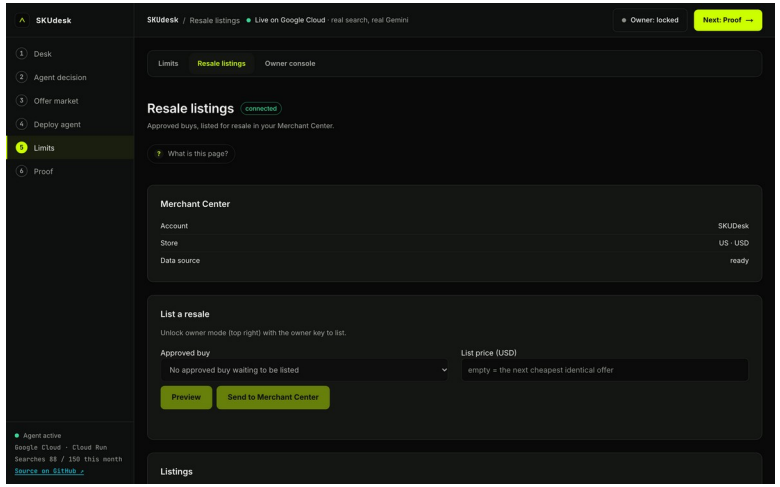












Market page (iPhone 16 Pro, LEGO Ferrari) · Resale listings · Run metrics · Landing · Approvals — live at skudesk.lol

Prototype performance report / benchmarking

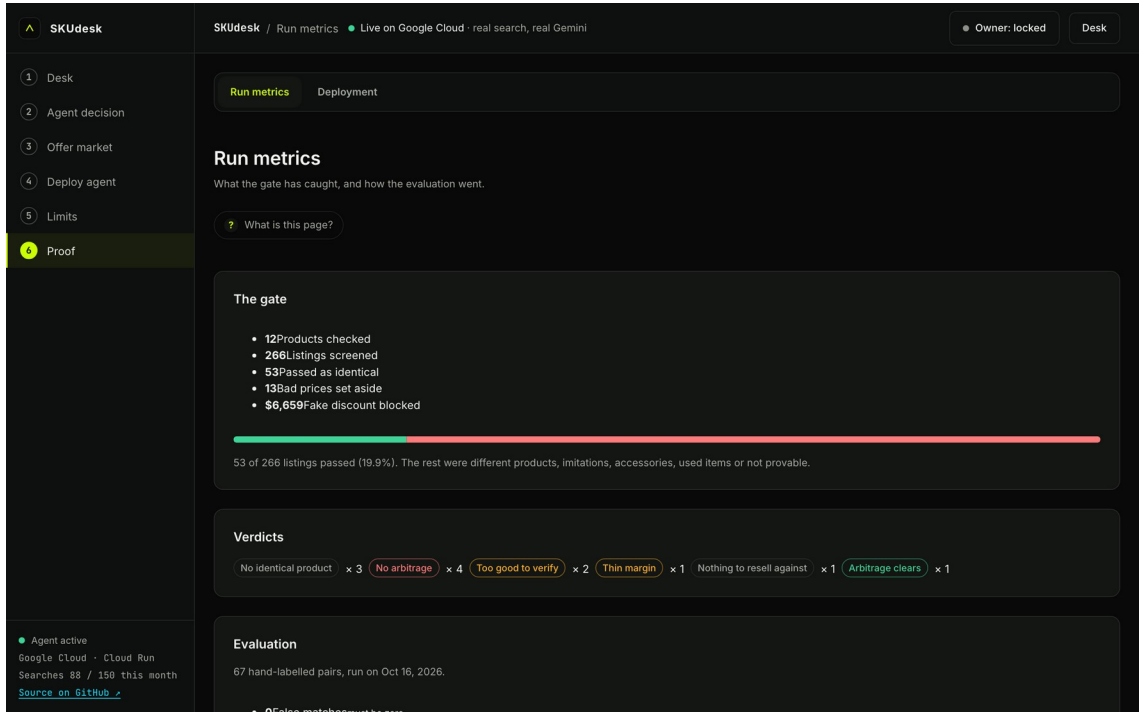
67 hand-labelled product pairs run against the real model: phones, LEGO sets, game consoles, accessories, clones, bundles, regional variants, injected instructions, Thai-language listings.

0 wrong matches (a different product called identical).

0 missed identical products.

64 of 67 exact on the verdict; the other three are “not provable”, which is the safe side.

Live Every check on the site adds to the run metrics: listings screened, passed as identical, bad prices set aside.



Additional details / future development

- Add eBay Browse as a second source (the adapter is built and joins when its key is added).
- Sold-price data for resale pricing, in place of asking prices.
- More categories beyond phones, devices and LEGO, each with its own identity rules and labelled pairs.
- Seller-side agents that bid with their own cost models on the order book.
- Alerts when a watched product opens a gap, with the approval link in the message.

Provide links to your:

- **GitHub public repository** https://github.com/johnnyduo/SKUdesk_
- **Demo video link (3 minutes)** https://youtu.be/DgpX9r0xt9Y
- **Final product link** https://skudesk.lol

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

